

CH13.1 空間中的平面方程式

$$1. 12x - 17y + 4z = 3 \quad 2. (1) \quad 1 \quad (2) \quad -\frac{1}{2} \quad 3. \pm \frac{\sqrt{2}}{3} \quad 4. \frac{2\sqrt{6}}{3} \quad 5. 5:4$$

$$6. 4x + 3y + z = 0 \quad 7. (1) \quad 27 \quad (2) \quad 3x + 6y + 2z = 18 \quad 8. \quad 1 \quad 9. x - 4y - 3z + 4 = 0$$

$$10. 15x - 10y + 5z = 78$$

解析

$$1. \overrightarrow{AB} = (-1, 0, 3), \overrightarrow{n_E} = (7, 4, -4)$$

$$\therefore \vec{n} = \left(\begin{vmatrix} 0 & 3 \\ 4 & -4 \end{vmatrix}, \begin{vmatrix} 3 & -1 \\ -4 & 7 \end{vmatrix}, \begin{vmatrix} -1 & 0 \\ 7 & 4 \end{vmatrix} \right)$$

$$= (-12, 17, -4) // (12, -17, 4)$$

令平面 $E': 12x - 17y + 4z = k$ ，將 $A(2, 1, -1)$ 代入 E'

$$24 - 17 - 4 = 3 = k$$

$$\therefore E': \underline{12x} - \underline{17y} + 4z = 3$$

$$2. (1) \because E_1 // E_2 \Rightarrow \frac{1}{1} = \frac{k}{1} = \frac{1}{k} \Rightarrow k = \underline{1}$$

$$(2) \because E_1 \perp E_2 \Rightarrow (1, k, 1) \cdot (1, 1, k) = 0$$

$$\Rightarrow 1 + k + k = 0 \Rightarrow k = -\frac{1}{2}$$

$$3. \overrightarrow{DA} = (-1, -2, -3), \overrightarrow{DB} = (-4, -5, -6)$$

$$\overrightarrow{DC} = (-2, -1, -3)$$

$\therefore$ 平面 DAB 法向量 $\vec{n}_1$

$$= \left(\begin{vmatrix} -2 & -3 \\ -5 & -6 \end{vmatrix}, \begin{vmatrix} -3 & -1 \\ -6 & -4 \end{vmatrix}, \begin{vmatrix} -1 & -2 \\ -4 & -5 \end{vmatrix} \right)$$

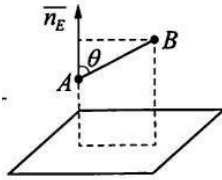
$$= (-3, 6, -3) // (1, -2, 1)$$

平面 DAC 法向量 $\vec{n}_2$

$$= \left(\begin{vmatrix} -2 & -3 \\ -1 & -3 \end{vmatrix}, \begin{vmatrix} -3 & -1 \\ -3 & -2 \end{vmatrix}, \begin{vmatrix} -1 & -2 \\ -2 & -1 \end{vmatrix} \right)$$

$$= (3, 3, -3) // (1, 1, -1)$$

$$\therefore \cos\theta = \pm \frac{1-2-1}{\sqrt{1+4+1} \cdot \sqrt{1+1+1}} = \pm \frac{2}{3\sqrt{2}} = \pm \frac{\sqrt{2}}{3}$$



$$4. \overrightarrow{AB} = (1, 1, 1), \overrightarrow{n_E} = (1, -1, 1)$$

$$\Rightarrow \cos\theta = \frac{(1, 1, 1) \cdot (1, -1, 1)}{\sqrt{3} \times \sqrt{3}} = \frac{1}{3}$$

$$\Rightarrow \sin\theta = \frac{2\sqrt{2}}{3}$$

$$\text{故 } \overline{AB} \text{ 在 } x - y + z = 1 \text{ 之正射影長} = |\overline{AB}| \sin\theta = \frac{2\sqrt{6}}{3}$$

5. $A(-2, -2, 2)$ 代入平面 E 得 $-4 - 2 - 4 - 5 < 0$ $B(6, 1, -2)$ 代入平面 E 得 $12 + 1 + 4 - 5 > 0$ 故 A, B 在平面異側

$$\begin{aligned} \overline{AC} : \overline{BC} &= d(A, E) : d(B, E) \\ &= \frac{|-4 - 2 - 4 - 5|}{\sqrt{4 + 1 + 4}} : \frac{|12 + 1 + 4 - 5|}{\sqrt{4 + 1 + 4}} \\ &= 15 : 12 = 5 : 4 \end{aligned}$$

$$6. \frac{|x+2y+3z+1|}{\sqrt{1+4+9}} = \frac{|3x+y-2z-1|}{\sqrt{9+1+4}}$$

$$\Leftrightarrow x + 2y + 3z + 1 = \pm(3x + y - 2z - 1)$$

$$\Rightarrow E_1: 2x - y - 5z - 2 = 0 \text{ 或 } E_2: 4x + 3y + z = 0$$

取 $x + 2y + 3z + 1 = 0$ 上一點 $A(-1, 0, 0)$

$$d_1 = d(A, E_1) = \frac{|-2 - 2|}{\sqrt{4 + 1 + 1}} = \frac{4}{\sqrt{6}}$$

$$d_2 = d(A, E_2) = \frac{|-4|}{\sqrt{16 + 9 + 1}} = \frac{4}{\sqrt{26}}$$

$\because d_2 < d_1 \therefore E_2: \underline{4x + 3y + z = 0}$ 為銳角平分面

7. (1) 令 $E: \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, 其中 $a > 0$, $b > 0$, $c > 0 \therefore \frac{2}{a} + \frac{1}{b} + \frac{3}{c} = 1$

由算幾不等式知

$$\frac{2}{a} + \frac{1}{b} + \frac{3}{c} \geq 3 \sqrt[3]{\frac{2}{a} \cdot \frac{1}{b} \cdot \frac{3}{c}}$$

$$\therefore \frac{1}{27} \geq \frac{6}{abc} \Rightarrow \frac{abc}{6} \geq 27 \therefore \text{最小值 } k = \underline{27}$$

$$(2) \text{等號成立於 } \frac{2}{a} = \frac{1}{b} = \frac{3}{c} = \frac{1}{3}$$

$$\Rightarrow a = 6, b = 3, c = 9$$

$$\therefore \text{平面 } E \text{ 方程式為 } \frac{x}{6} + \frac{y}{3} + \frac{z}{9} = 1 \Rightarrow \underline{3x + 6y + 2z = 18}$$

$$8. \overrightarrow{PQ} = (2, 2, -9), \overrightarrow{PR} = (-2, 3, -6)$$

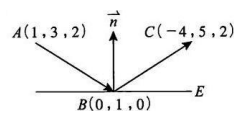
$$\begin{aligned} \overrightarrow{PQ} \times \overrightarrow{PR} &= \left(\begin{vmatrix} 2 & -9 \\ 3 & -6 \end{vmatrix}, \begin{vmatrix} -9 & 2 \\ -6 & -2 \end{vmatrix}, \begin{vmatrix} 2 & 2 \\ -2 & 3 \end{vmatrix} \right) \\ &= (15, 30, 10) // (3, 6, 2) \end{aligned}$$

平面 PQR 的方程式為

$$3(x - 0) + 6(y + 1) + 2(z - 6) = 0 \Rightarrow 3x + 6y + 2z = 6$$

$$\therefore A(2, 0, 0), B(0, 1, 0), C(0, 0, 3)$$

四面體 $O - ABC$ 之體積為 $\frac{1}{6} \times 2 \times 1 \times 3 = \underline{1}$



$$9. \overrightarrow{BA} = (1, 2, 2)$$

$$\overrightarrow{BC} = (-4, 4, 2)$$

如右圖

$$\text{令 } \vec{n} = \frac{\overrightarrow{BA}}{|\overrightarrow{BA}|} + \frac{\overrightarrow{BC}}{|\overrightarrow{BC}|} = \frac{(1,2,2)}{\sqrt{1+4+4}} + \frac{(-4,4,2)}{\sqrt{16+16+4}} = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right) + \left(\frac{-4}{6}, \frac{4}{6}, \frac{2}{6}\right) = \left(\frac{-1}{3}, \frac{4}{3}, 1\right)$$

$$\text{平面 } E: \frac{-1}{3}(x-0) + \frac{4}{3}(y-1) + 1(z-0) = 0$$

$$\Leftrightarrow \underline{x - 4y - 3z + 4 = 0}$$

10. 設平面方程式為 $3x - 2y + z = k$

$$\text{則 } \frac{k}{3} + \frac{k}{-2} + k = 13 \Rightarrow 2k - 3k + 6k = 78$$

$$\text{得 } k = \frac{78}{5}$$

$\therefore$ 平面方程式為 $15x - 10y + 5z = 78$